📋 PROJECT BRIEF: Tolveris Launchpad

Team Name: Team Tolveris
Hackathon Track: Payment & Consumer Applications
Document Purpose: Core Alignment, System Architecture, and Development Roadmap

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1. Executive Summary

Project Overview

Tolveris Launchpad is a blockchain-powered financial safety platform designed to help senior citizens and financially vulnerable Filipinos safely adopt digital banking through financial literacy, intelligent scam prevention, and accessible digital savings.

Unlike conventional banking applications that primarily focus on transactions, Tolveris emphasizes **financial protection before financial participation**. Users are guided through educational modules, encouraged to develop healthy saving habits through an "Ipon" savings system, and protected by an AI-powered transaction assistant that identifies potentially suspicious activities before money is transferred.

The platform is designed to leverage the **Stellar blockchain** as its production infrastructure, enabling secure digital payments, transparent transaction records, and future interoperability with existing financial institutions.

For the hackathon, the blockchain layer will be represented by a **production-like blockchain simulation** running inside a Virtual Machine (VM). This environment exists solely during the development phase to validate the application's architecture, transaction flow, and blockchain interactions before migrating to a live Stellar deployment.

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2. Problem Statement

The Growing Risk of Digital Financial Scams

The Philippines has rapidly adopted digital financial services through platforms such as GCash, Maya, online banking, and digital payment systems. While these technologies have increased accessibility and convenience, they have also introduced new risks for users with limited digital literacy.

Senior citizens and financially vulnerable individuals are among the most common victims of:

- Phishing attacks
- Fake banking websites
- OTP scams
- Investment fraud
- Social engineering attacks
- Fake payment requests
- Impersonation scams

Many users struggle to distinguish legitimate financial transactions from malicious ones, often resulting in financial loss.

Current banking applications assume users already possess sufficient digital knowledge, leaving little room for education or guided decision-making.

As digital banking continues to expand, there is an increasing need for platforms that prioritize user protection, accessibility, and financial literacy.

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3. Vision

Tolveris envisions a financial ecosystem where every user—regardless of age or technical experience—can confidently participate in digital banking without fear of scams or financial exploitation.

By combining:

- Financial literacy
- Accessible digital banking
- Intelligent transaction guidance
- Secure blockchain infrastructure
- Transparent financial records

the platform empowers users to develop safe financial habits while maintaining ownership and confidence over their digital assets.

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4. User Personas

Primary Persona — Senior Citizens

Objective

Help elderly users safely transition into digital banking through education and intelligent guidance.

The platform focuses on:

- Easy-to-understand banking workflows
- Simple savings management
- Scam awareness
- Safe transaction verification
- Financial literacy

Rather than overwhelming users with advanced investment products or complicated financial dashboards, Tolveris prioritizes clarity, accessibility, and confidence.

Secondary Persona — Researchers and Data Scientists

Tolveris also creates value beyond consumer banking.

Because blockchain transactions are publicly verifiable while preserving user anonymity, they can become valuable datasets for researchers studying financial behavior and fraud detection.

Potential research applications include:

- Scam detection research
- Fraud analytics
- Transaction graph analysis
- Behavioral finance
- Financial inclusion studies
- Machine learning datasets
- Anomaly detection

Researchers may develop their own analytical models using publicly available blockchain transaction data without exposing personally identifiable information.

Tolveris itself demonstrates one practical implementation through its proprietary transaction risk assessment model.

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5. Core Features

A. Safe Digital Banking

Tolveris provides a simplified banking interface designed specifically for accessibility.

Features include:

- Digital wallet
- Account balance
- Deposit and withdrawal functionality
- Money transfers
- Transaction history
- Large buttons and readable typography
- Minimal and intuitive navigation

The interface is intentionally designed to reduce confusion and cognitive load for elderly users while maintaining familiar banking workflows.

B. Ipon Savings System

Instead of focusing on traditional budgeting, Tolveris encourages users to build healthy financial habits through goal-oriented saving.

Users can create multiple savings goals, including:

- Emergency Fund
- Medicine Fund
- Retirement Savings
- Family Savings
- Holiday Savings
- Education Savings

Each savings goal displays:

- Progress percentage
- Total saved amount
- Deposit history
- Goal completion milestones

The objective is to encourage consistent saving behavior without requiring users to manually categorize every expense.

C. Financial Literacy Modules

Education is integrated directly into the banking experience.

Before unlocking certain features, users complete short, easy-to-understand learning modules covering topics such as:

- Introduction to Digital Banking
- Protecting Your OTP
- Recognizing Phishing Attempts
- Safe Online Transactions
- Identifying Investment Scams
- Responsible Saving
- Digital Wallet Security

Each lesson is designed to take approximately one to two minutes to complete.

After completing a module, users receive a blockchain-issued **Financial Literacy Badge**, providing a permanent and verifiable record of their educational achievements.

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6. Intelligent Scam Prevention

Protecting users from scams is one of the platform's primary objectives.

Rather than preventing every transaction automatically, Tolveris helps users understand why a transaction may be risky before proceeding.

Transaction Risk Analysis

Every outgoing transaction is evaluated using multiple indicators, including:

- Recipient familiarity
- Transaction amount
- Previous account activity
- Transfer frequency
- Behavioral anomalies
- Rule-based fraud indicators

The system analyzes these characteristics to determine whether a transaction appears normal or potentially suspicious.

When elevated risk is detected, users receive an explanation before the transaction proceeds.

Example:

> **Warning:** This transaction appears unusual because you have never transferred funds to this recipient before. Please verify the recipient before continuing.

The goal is to educate users rather than simply interrupt them.

AI-Powered Transaction Assistant

Tolveris includes an AI-powered transaction assistant that helps users make informed financial decisions.

The assistant is supported by Team Tolveris' **proprietary transaction risk assessment model**, which evaluates transaction context and classifies activities based on their likelihood of being legitimate or potentially malicious.

The model considers factors such as:

- Transaction patterns
- Recipient history
- User behavior
- Transaction frequency
- Amount anomalies
- Rule-based fraud indicators

Instead of automatically blocking transactions, the assistant explains potential risks in clear and accessible language that non-technical users can easily understand.

For example:

> "This transfer is significantly larger than your usual transactions and is being sent to a new recipient. Please verify that you personally know this individual before continuing."

The assistant serves as an educational companion that empowers users to recognize warning signs, understand potential risks, and make safer financial decisions while maintaining full control over their accounts.

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7. Blockchain Architecture

Production Architecture

Tolveris is designed to operate on the **Stellar blockchain**, a decentralized network built for secure, fast, and low-cost financial transactions.

By leveraging Stellar, the platform can provide users with transparent financial records while maintaining the security and efficiency expected from modern digital banking systems.

The production implementation is planned to include:

- Stellar Wallets
- Stellar Ledger
- Soroban Smart Contracts
- USDC Stablecoin Support
- SEP-24 / SEP-6 Anchor Integrations
- Blockchain-issued Financial Literacy Badges

Using Stellar provides several important advantages:

Transparency

Every transaction is permanently recorded on an immutable public ledger, allowing financial records to remain auditable and trustworthy without exposing sensitive personal information.

Security

Blockchain technology helps ensure transaction integrity by preventing unauthorized modification of transaction records and providing cryptographic verification of account ownership.

Low Transaction Costs

Stellar is designed for efficient value transfers with minimal network fees, making it practical for everyday consumer banking applications.

Fast Settlement

Transactions can be confirmed within seconds, allowing users to transfer funds quickly while maintaining a reliable and secure payment experience.

Verifiable Digital Credentials

Financial Literacy Badges issued on-chain become permanent digital credentials that users can present as verifiable proof of completing financial education programs.

By combining these capabilities, Stellar provides the technical foundation for a secure, scalable, and transparent financial safety platform.

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8. Development Architecture

Production-Like Blockchain Simulation

While the long-term vision of Tolveris is a full deployment on the Stellar blockchain, the hackathon prototype will utilize a **production-like blockchain simulation** during development.

This simulation exists **only as a proof of concept** and is **not** intended to replace the final blockchain implementation.

Instead, it mirrors the expected behavior of the production environment, allowing the team to validate application logic, wallet interactions, transaction processing, and blockchain communication before migrating to Stellar.

The development environment includes:

- Production-like blockchain network simulation
- Virtual Machine development environment
- Simulated Stellar-compatible wallets
- Simulated distributed ledger
- Simulated transaction validation
- Simulated blockchain state management

Because the simulation closely reflects the intended production architecture, the migration path primarily involves replacing the simulated blockchain services with their corresponding Stellar components while preserving the application's core business logic.

This development strategy allows the team to demonstrate a complete end-to-end workflow during the hackathon while maintaining a realistic roadmap toward production deployment.

9. System Workflow

The following workflow demonstrates the intended user journey within Tolveris, from onboarding to secure digital banking.

Step 1 — Account Registration

The user creates a Tolveris account and is provided with a blockchain wallet that will be used for future digital banking activities.

Step 2 — Financial Literacy Onboarding

Before accessing advanced banking features, users complete short educational modules covering digital banking fundamentals, online safety, and scam awareness.

Upon completion, the system issues a **Financial Literacy Badge**, which is permanently recorded on the blockchain.

Step 3 — Ipon Savings Setup

Users create one or more savings goals based on their personal needs.

Examples include:

- Emergency Fund
- Medicine Fund
- Retirement Savings
- Family Savings
- Holiday Savings

The application continuously tracks savings progress and encourages healthy financial habits.

Step 4 — Digital Banking

Users perform everyday banking activities such as:

- Depositing funds
- Viewing balances
- Sending money
- Reviewing transaction history

Every completed transaction is securely recorded on the blockchain ledger.

Step 5 — Intelligent Transaction Review

Whenever a transfer is initiated, the transaction is analyzed by the platform's transaction risk assessment model.

If suspicious behavior is detected, the system presents contextual guidance explaining why the transfer may require additional verification before the user proceeds.

Step 6 — Secure Transaction Recording

Once the user confirms the transaction, the transfer is finalized and recorded on the blockchain ledger, creating a transparent and tamper-resistant financial record.

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10. Demonstration Scenarios

Scenario 1 — Senior Citizen Banking Experience

Persona

Maria Santos

- 67 years old
- First-time digital banking user
- Limited experience with online financial services
- Wants a safe and simple way to save money

Demonstration Flow

Maria creates her account and receives a digital wallet.

She completes the introductory financial literacy lesson explaining phishing scams and OTP protection.

After finishing the lesson, she earns a Financial Literacy Badge.

Maria creates a **Medicine Savings Goal** and deposits funds into her account.

Later, she receives a fraudulent message asking her to transfer money to an unfamiliar account.

Before completing the transfer, Tolveris analyzes the transaction and displays a warning explaining that the recipient is unfamiliar and the activity appears unusual.

Maria recognizes the warning signs and cancels the transaction.

This demonstration highlights how education and intelligent transaction guidance can reduce the likelihood of financial scams.

Scenario 2 — Research and Analytics

Persona

University Researcher

The researcher studies anonymized blockchain transaction records to understand financial behaviors and identify patterns associated with fraudulent activity.

Because blockchain records are transparent while preserving user privacy, researchers can analyze transaction behavior without exposing personal identities.

Possible research applications include:

- Fraud detection
- Transaction anomaly detection
- Behavioral finance
- Digital payment trends
- Financial inclusion research
- Machine learning model development

Tolveris demonstrates one implementation through its own transaction risk assessment model, while enabling future researchers to develop and evaluate their own approaches using publicly available blockchain data.

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11. Technical Stack

Frontend

- React
- Next.js
- Tailwind CSS

Backend

- Python (FastAPI) or Node.js (Express)

Database

- PostgreSQL
- SQLite (Development)

AI & Intelligent Analysis

- Proprietary transaction risk assessment model
- Transaction behavior analysis
- Rule-based fraud detection
- Risk classification engine
- Explainable AI transaction guidance

Blockchain

Development Environment

- Production-like blockchain simulation

- Virtual Machine development environment
- Simulated Stellar-compatible wallets
- Simulated distributed ledger
- Simulated blockchain state management
- Simulated transaction validation

Production Environment

- Stellar Network
- Stellar Wallets
- Stellar Ledger
- Soroban Smart Contracts
- USDC Stablecoin
- SEP-24 / SEP-6 Anchor Integrations
- Blockchain-issued Financial Literacy Badges

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12. Hackathon Strategy

Tolveris is designed with the understanding that this hackathon emphasizes **innovation, feasibility, and presentation** as much as technical implementation.

Rather than attempting to build every production component within a limited timeframe, the team will focus on delivering a polished proof of concept that clearly demonstrates the platform's value, architecture, and future scalability.

The project emphasizes:

- A polished and accessible user interface.
- A complete end-to-end user journey.
- A believable blockchain architecture.
- Intelligent scam prevention.
- Financial literacy integration.
- A realistic migration path from development simulation to production deployment.

By using a production-like blockchain simulation during development, the team can validate the proposed architecture while demonstrating how the application will operate once connected to the live Stellar network.

This approach allows development effort to focus on user experience, product functionality, and architectural credibility instead of infrastructure that would otherwise require significantly more time and resources.

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13. Future Roadmap

Following the hackathon, Tolveris is intended to evolve into a production-ready platform through a phased migration strategy.

Phase 1 — Production Blockchain Deployment

- Replace the simulated blockchain with the Stellar Network.
- Deploy real Stellar wallets.
- Record transactions directly on-chain.
- Implement Soroban smart contracts.

Phase 2 — Financial Services Integration

- Integrate USDC for digital savings.
- Connect with licensed Stellar Anchors through SEP-24 and SEP-6.
- Enable interoperability with Philippine digital payment providers where applicable.

Phase 3 — Advanced Security

- Expand the transaction risk assessment model.
- Introduce adaptive fraud detection.
- Improve explainability of transaction guidance.
- Add stronger authentication and account recovery mechanisms.

Phase 4 — Accessibility Expansion

- Voice-assisted navigation.
- Multi-language support.
- Accessibility enhancements for senior citizens.
- Family assistance and trusted contact features.

Phase 5 — Research Ecosystem

- Publish anonymized datasets for academic use.
- Support fraud detection benchmarking.
- Encourage universities and researchers to develop independent financial risk models using blockchain transaction data.

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14. Expected Impact

Tolveris seeks to create meaningful social and technological impact by making digital banking safer and more accessible.

The platform aims to:

- Reduce financial scams targeting senior citizens.
- Improve financial literacy through guided education.
- Encourage consistent saving habits through the Ipon system.
- Increase confidence in digital banking.
- Demonstrate responsible blockchain adoption for consumer applications.
- Promote transparency through immutable blockchain records.
- Support academic research using anonymized blockchain transaction data.
- Showcase practical AI-assisted financial guidance for everyday users.

By focusing on education, accessibility, and security, Tolveris contributes to a more inclusive and trustworthy digital financial ecosystem.

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15. Team Roles

Dawn Andrei Pamesa

Project Lead & System Architect

Responsibilities:

- Overall project direction
- System architecture
- Product strategy
- Database design
- Blockchain migration planning
- Documentation
- Presentation and pitching

John Andrew Balbarosa

Blockchain & Infrastructure Engineer

Responsibilities:

- Blockchain architecture
- Development simulation environment
- Stellar integration planning
- Wallet management
- Transaction infrastructure

Colin James Daradar

Backend & Full-Stack Developer

Responsibilities:

- Backend APIs
- Banking system implementation
- User management
- Savings system
- Database integration

Mark Diaz

Security & AI Systems Engineer

Responsibilities:

- Transaction risk assessment model
- Scam detection logic
- AI-assisted transaction guidance
- Security architecture
- Financial Literacy Badge implementation

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16. Conclusion

Tolveris demonstrates how blockchain technology can be applied beyond cryptocurrency by addressing a practical social problem: protecting vulnerable users as they transition into digital finance.

Through accessible banking, guided financial education, intelligent transaction analysis, and transparent blockchain infrastructure, the platform encourages safer financial behavior while maintaining a clear path toward real-world deployment on the Stellar Network.

Although the hackathon implementation utilizes a production-like blockchain simulation for development and validation, the long-term vision remains a fully blockchain-powered financial safety platform capable of supporting consumers, researchers, and future financial innovations.

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17. One-Sentence Pitch

Tolveris Launchpad is a blockchain-powered financial safety platform that empowers senior citizens to bank with confidence through financial literacy, intelligent scam prevention, and secure digital savings, while leveraging Stellar's blockchain infrastructure and anonymized transaction data to advance transparency, financial inclusion, and fraud research.